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Challenges and opportunities of sharing animal health data for research and disease management: a case study of bovine tuberculosis

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Summary

The sharing of animal disease data should be encouraged. The analysis of such data will broaden our knowledge of animal diseases and potentially provide insights into their management. However, the need to conform to data protection rules in the sharing of such data for analysis purposes often poses practical difficulties.

This paper sets out the challenges and the methods used for the sharing of animal health data in England, Scotland and Wales – Great Britain – using bovine tuberculosis (bTB) data as a case study. The data sharing described is undertaken by the Animal and Plant Health Agency on behalf of the Department for Environment, Food and Rural Affairs and the Welsh and Scottish Governments. It should be noted that animal health data are held at the level of Great Britain (rather than the United Kingdom – which includes Northern Ireland), as Northern Ireland's Department of Agriculture, Environment and Rural Affairs has its own separate data systems.

Bovine tuberculosis is the most significant and costly animal health problem facing cattle farmers in England and Wales. It can be devastating for farmers and farming communities and the control costs for taxpayers in Great Britain are over £150 million a year.

The authors describe two methods of data sharing – first, where data are requested by, and delivered to, an academic institution for epidemiological or scientific analysis, and second, where data are proactively published in an accessible and meaningful way. They provide details of an example of the second method, namely, the free-to-access website 'information bovine TB' (https://ibtb.co.uk), which publishes bTB data for the benefit of the farming community and veterinary health professionals.

Keywords

Animal health data – Bovine tuberculosis – Data sharing – Disease management – ibTB – Information bovine tuberculosis – UK GDPR – United Kingdom General Data Protection Regulation.

Background

The advantages to be gained from the sharing of data have long been recognised. In 2013, the Department for Environment, Food and Rural Affairs (Defra) – as part of a wider United Kingdom (UK) Government initiative – published its 'Open Data Strategy' [1]. One of the key principles of this initiative was that data should be open by default. The hope across Government was that such a strategy, delivering access to Government-held data for both individuals and companies, would facilitate a host of both scientific and business opportunities. Defra later made a commitment to publishing 8,000 data sets as 'open data' by June 2016 under the initiative #OpenDefra.

Specifically, in regard to bovine tuberculosis (bTB), an independent review of Defra's TB strategy in England [2] concluded that industry must take greater responsibility for on-farm controls, biosecurity and safe trading and that Government needs to supply stakeholders with the relevant TB information to facilitate that.

However, the wish for increased accessibility to data has to be weighed against the requirement to adhere to data protection considerations. In the European Union (EU), these considerations are covered by the General Data Protection Regulation (GDPR), and although the Regulation has not applied in the UK since its departure from the EU, its provisions have been retained in UK law (the Data Protection Act 2018 [3]) as the UK GDPR. The aim of the UK GDPR is to give individuals (data subjects) more control over their personal data and its processing. It sets out a number of key requirements, which include requiring the consent of subjects for data sharing and processing and the anonymising of collected data to protect privacy. The implications of failing to comply with the UK GDPR can be serious; in recent years, fines in the millions of pounds have been handed out to companies and individuals who have failed to comply.

The data provided to researchers must conform to the principles of the UK GDPR, which include 'purpose limitation', i.e. researchers must have a clear intended use for the data, and 'data minimisation', i.e. they should only be provided with the data necessary to service that use. The practical implication of the latter is that the 'data owner' cannot simply hand over all data and leave the task of extracting what is required to the requestor, but must take appropriate steps to ensure only the data required are shared.

The wish to both increase accessibility to data and adhere to the UK GDPR has the potential to pull an organisation in opposite directions. Indeed, there is a danger that a strict interpretation of the UK GDPR by organisations (including Government) could stifle access to its data and the consequent benefits to be gained from any analysis.

Since the UK GDPR relates to the collection and processing of personal information, the first consideration when looking to share data is to decide whether the data to be shared include personal information. If they do, then the restrictions required by the UK GDPR apply. Personal data are defined as information that relates to an identified or identifiable individual. The data may be obviously and directly of a personal nature, e.g. name and address, or indirectly personal, meaning the data could be used to identify an individual.

In the main, scientific researchers are not interested in direct personal information, and even when they are, research ethics committees place limitations around researchers accessing personal information, to ensure that any research participants are anonymised. However, researchers may require data (e.g. a farm or animal identifier) that could potentially be linked to personal data, i.e. is indirectly personal data. The decision as to whether such data are indirectly personal is often not straightforward and will need to be decided on a case-by-case basis.

In general, direct personal data are only required for projects such as socio-economic surveys or the identification of farms for on-farm studies. These types of studies are relatively rare and are far outnumbered by epidemiological studies, which do not require direct personal and confidential data.

Where data are deemed to be personal (either directly or indirectly), the UK GDPR places restrictions on how they are shared, e.g. they cannot be shared without obtaining the prior consent of the data subject. In general, obtaining prior consent in the case of bTB would be unduly onerous, as it would mean contacting thousands of herd owners – there are currently bTB testing records in Great Britain for more than 71,000 'live' herds, as well as records on herds that no longer exist.

However, if the data are considered to be in the public interest, they are exempt from the requirements of the UK GDPR and can be shared without the consent of the data subject. The use of personal data in research and development that help to protect public and animal health is normally seen as compatible with the UK GDPR public interest exemption. This is the legal basis on which the Animal and Plant Health Agency (APHA) shares key animal health data that are deemed personal.

In addition to the requirement to adhere to the UK GDPR, the sharing and publication of bTB data are also covered in the Tuberculosis in Animals (England) Order 2021 [4] – 'The TB Order'. The Order sets out the rules and regulations relating to the control of this notifiable disease. With regards to data sharing and the publication of information, Section 23 of the Order states that:

> '(1) The Secretary of State may publish information regarding how long a bovine herd has had officially

tuberculosis-free status for the purpose of helping other persons to protect against the further spread of tuberculosis.

(2) Where a bovine herd loses its officially tuberculosis-free status the Secretary of State may publish information regarding that herd for the purpose of helping other persons to protect against the further spread of tuberculosis.

(3) Information which may be published under paragraph (1) or (2) may be published in any form that the Secretary of State sees fit.'

As can be seen from the above, the TB Order allows the publication of any and all data related to a herd's TB status, in any format, if it is thought that the data support the public interest task of reducing the spread of infection.

Bovine tuberculosis data

Of all the diseases dealt with by APHA, bTB is associated with the most abundant and comprehensive data. It is a notifiable disease with a compulsory cattle testing regime in place. This testing, which is carried out by APHA (or on its behalf), has been electronically recorded since the mid-1980s, meaning that an enormous amount of data has built up and is available for analysis.

Bovine TB can be considered a highly 'politicised' disease, with entrenched views as to its causes and how it should be managed. This makes the need for data sharing to be transparent and equitable very important, possibly more so than for any other animal health disease in Great Britain.

Although the relational database holding this information consists of over 30 tables, from an epidemiological point of view the main data tables are:

- **Herd**: Contains data relating to the cattle herd being tested, e.g. its location and type (beef, dairy, etc.). Currently there are approximately 240,000 herd records. It should be noted that only approximately 71,000 of these are current 'live' herds in Great Britain; the remainder of these records relate to 'archived' herds (herds that no longer exist).

- **Test**: Contains data relating to the testing of a herd (or sub-set of it) for bTB. Tests fall into two categories: firstly, surveillance tests, the frequency of which is primarily determined by the prevalence of bTB in the location of the herd (currently either every six months, one year or four years) and secondly, control tests carried out after bTB is identified or suspected. These tests must be negative before the restrictions associated with bTB occurrence are lifted. Currently there are approximately three million testing records from 1986 onward.

- **Breakdown**: Data relating to the bTB herd incident (or 'breakdown') that are recorded once bTB positive animals are found (or suspected) in a herd. This would include, for example, information on the start and end of the movement restrictions that are imposed on such herds. Currently there are approximately 98,000 breakdown records from 1986 onward.

- **Animal**: Details of all cattle tested, e.g. sex and age. Up until 2012, only the details of animals that tested positive were recorded, but since that time details for all animals subject to a test have been recorded. For ease of processing, these data are split into two data sets, one with test-positive animals (currently approximately 1.5 million records from 1986 onward) and one with test-negative animals (approximately 95 million from 2012 onward).

Although the bTB testing data are the main data set relating to bTB in Great Britain, it is only one of many. An example of another major bTB-related data set is from the Randomised Badger Culling Trial (RBCT). The RBCT was a large study looking at the effect of badger culling on cattle bTB incidence in which the study treatments were carried out between 1999 and 2005. In common with other research or trial-based studies, these data were not made available for data sharing until the study had concluded and relevant peer-reviewed papers had been published. In general, once studies have been quality assured through peer review, Defra and the devolved administrations are keen to maximise their value by making the data available for further analysis.

Although not specifically a bTB data set, the Cattle Tracing System (CTS) data are often used in tandem with bTB data to analyse the effect of cattle movement on bTB incidence. Originally developed in response to the bovine spongiform encephalopathy crisis [5], the CTS contains the birth and death records of all cattle, along with details of every movement to or from farms and markets and to slaughter. The CTS data therefore represent a remarkable store of data on cattle movements in Great Britain. The CTS currently contains data on approximately 300 million individual movements of animals (dating back to 2001, when the recording of cattle movements became compulsory). Although the primary reason for the recording of cattle movements is for regulatory and tracing purposes, the accumulated data set represents an extremely valuable epidemiological and research asset.

Sharing data

Data requestors cover a wide spectrum both within government and externally. Internal requests (from within Defra and its associated agencies) come from epidemiologists,

statisticians and policy-makers working in the field of bTB. External requests come from a variety of sources, ranging from students requiring bTB data for undergraduate projects to consortia of university research departments carrying out three-year research projects.

In general, external requests (i.e. requests from outside the government department holding the data) will require a data sharing agreement to be in place. For such requests, APHA liaises with the requestor about its data requirements and creates a request specification that can be inserted into a confidentiality agreement. If, however, the request is made under the Freedom of Information (FOI) Act [6], the request must be dealt with exactly as set out in the request and there is no interaction with the requestor. Because of this, APHA would recommend that FOI requests are not an ideal way of obtaining scientific research data.

Once a draft data sharing agreement has been created, approval is required from the Information Asset Owner (the person responsible for the information asset in the organisation that holds the data). This is likely to be more than one person if multiple data sets are requested. Permission to share data is also required from a person appointed by the Scottish or Welsh Government if data from these nations have been requested.

Once all the necessary approvals have been given, the data are securely sent to the requestor along with appropriate metadata. This usually involves uploading the files to a secure site and sending unique access details to the requestor separately. It should be noted that the data agreement specifies that data access is time-limited and that after a specified date the data should be destroyed. Furthermore, access is given on a 'per project basis', meaning that if access to the data is required for a different project, a new approval has to be given.

Over the past 20 years, the sharing of bTB data by APHA has enabled a wide variety of scientific research to be undertaken. This has ranged from traditional epidemiological, statistical, spatial and social science analysis to more novel approaches such as machine learning, deep learning and genetic analysis. The work involved in the creation of bespoke data sets for analysis should not be underestimated; together with the provision of expertise relating to the available data, it forms a vital part of any project. As such, those responsible for assembling the project data should be recognised in any peer-reviewed papers resulting from the work and the vast majority of lead authors are more than happy to include those responsible for data provision as co-authors.

The following examples illustrate the range of work that has utilised APHA's bTB data in recent years:

- Donnelly *et al*. [7] analysed the RBCT data to establish the effectiveness of badger culling in reducing the incidence of bTB in cattle.
- Gilbert *et al*. [8] used the data to definitively establish the link between cattle movement and bTB infection.
- Van Tonder *et al*. [9] have combined APHA-based genetic data with the testing data described above to better understand the relationship between badger and cattle infection.
- Banos *et al*. [10] have created an index of resistance to TB infection in dairy cattle ('TB Advantage'). This index is now used commercially and is a good example of the application of machine learning.
- Denholm *et al*. [11] have used deep-learning techniques to try and predict bTB occurrence from the spectral analysis of milk samples using deep learning analogy.

The majority of analyses combine a number of data sets; e.g. Donnelly *et al*. combined bTB testing data and RBCT data, and Gilbert *et al*. combined bTB testing data and CTS data. The ability to combine bTB data easily and accurately from different sources highlights the power of having data in a relational database format.

Publishing data

Defra has long published bTB breakdown numbers in an aggregate form as accredited national statistics [12]. Such data provide a narrative to the general public on progress in dealing with bTB at a national and county level and the bTB testing effort, but are not designed to provide details relating to individual incidents.

Since the early 2000s there has been a drive from Defra to encourage farmers to take greater 'ownership' of the bTB problem. To support that objective it introduced a series of measures to encourage them to introduce better on-farm biosecurity and to pursue 'informed' or 'knowledge-based' purchasing – i.e. to select animals for purchase taking the bTB history of the herd they came from into account. In general, both the institutions representing the farming community (e.g. farming unions) and Defra agreed that, to enable farmers to take these steps, better access to bTB data was required. Accordingly, Defra committed to publishing more detailed bTB information at the level of individual herds. This represented a radical change in thinking, as up until that point farmers' bTB history had been deemed personal information that should not be made public.

The first step in the process of publishing herd data was for Defra Ministers to introduce a new statutory power in the Tuberculosis (England) Order 2014 (since replaced by the Tuberculosis in Animals [England] Order 2021 referred to above). Once this had been achieved, it was decided that the most effective way to publish the herd-level data would be in an interactive spatial format. Consequently, the

Environmental Research Group Oxford, in partnership with APHA, was commissioned to create an online interactive map showing the locations of bTB breakdowns: 'information bovine TB' or 'ibTB' (https://ibtb.co.uk). The initial version of the system first went online in the summer of 2015. It displayed current ('live') incidents of bTB and historical incidents that had occurred in the previous five years (later increased to ten years). Figure 1 shows the ibTB start-up page displaying the 'live' incidents of bTB in England and Wales. From this initial view users can drill-down to their area of interest; as they do this, the clusters of incidents become smaller until (at the finest scale) only individual incidents are displayed.

Since its inception, a series of upgrades have been implemented – the latest of which went live on 31 January 2022. It allows users to create bespoke displays of farms that have been free of bTB for a given number of years. The number of years a herd has been bTB free is a good (and easy to communicate) proxy for bTB risk status [13]. This upgrade is only available for English herds, because the ability to publish this information has not yet been approved by the Welsh Government. Scotland has its own Officially Tuberculosis Free status and its bTB incidents have never formed part of the system. The ibTB system is also linked to the commercial trading app SellMyLivestock (https://sellmylivestock.com) to encourage farmers to take account of the bTB risk in purchases they make.

In a series of usability trials carried out in 2018, farmers and veterinarians using the system rated its usability as 'excellent' [14].

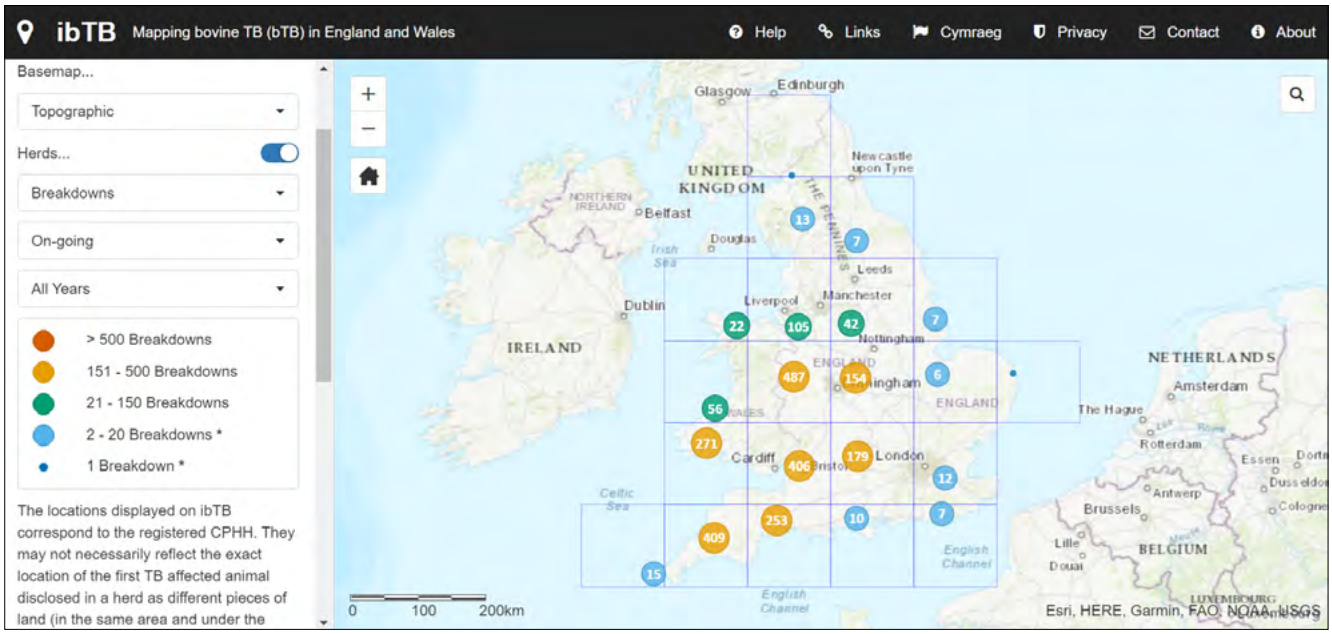
Since its inception, ibTB has been visited over 1 million times by 200 thousand unique users, showing that there is an appetite from the farming community for access to this type of information. The system is now seen as a key component of Defra's advice for minimising bTB risk [15].

Because ibTB is a free-access system that does not require users to register prior to use, it is not currently possible to know exactly who is using it. It is certainly being used by the farming community to better help farmers understand the disease picture in their area, as well as inform their purchasing decisions. But there have also been some unforeseen uses of the data – e.g. groups opposed to badger culling have used it as a basis for their analyses.

Discussion

The sharing of animal data in an efficient manner is crucial both for encouraging and enabling scientific research and for successful disease control. However, it is important that the privacy rights of the data subjects (farmers) are adhered to by conforming to UK GDPR specifications. Equally, such regulation should not become an impediment to data sharing where this is in the public interest.

Bovine TB in Great Britain has a wealth of good quality data associated with it going back over 30 years and these data have underpinned much high-quality and innovative research during that time. In more recent years they have been



CPHH: County Parish Holding Herd

Figure 1

The ibTB ('information bovine tuberculosis') display at start-up

The start-up page of the website shows live incidents of bovine tuberculosis in England and Wales and allows users to zoom in on areas of interest

published in an accessible and user-friendly way via ibTB to empower farmers to take action to protect their herds from the disease.

The ibTB system has proved to be an effective way to pro-actively publish bTB data in a meaningful way. In general, the reaction to the system has been positive, with only a few (under ten) farmers objecting to either Defra or APHA about their bTB history being made public. The system is of course limited by the availability and accuracy of bTB data. The locational data available are restricted to the location of the affected farm, and some farmers have made it known that they wish to see the data at a finer scale, e.g. the location of infected animals – data that are not currently collected.

The lack of detailed knowledge about the users of the ibTB system currently makes any meaningful evaluation of its impact impossible. If the users of the system were known, it would be possible to understand, for example, what impact (if any) it had on their trading decision-making. The only way to understand the user base would be to implement a registration system. Such a system has been suggested but has met with some resistance from both policy-makers and users, mainly on the basis that it is seen as a potential impediment to use. Even if such a registration system were implemented in the future, it is highly likely that it would be voluntary, which would limit its usefulness from a user analysis point of view.

Despite the substantial use that has been made of bTB data over the past 30 years, more can and should be done to encourage their use by making the research community and other interested parties more aware of what is available and making access to them easier and more efficient. The most obvious way of doing this would be to develop online access to metadata describing the data and make it possible for users to download those data that were non-personal and non-confidential. The creation of such an online portal for bTB data would be a significant piece of work, but the authors believe the effort to create it would pay dividends in the professional and public interest benefits it would bring. Another option (which is currently in progress) is the development of an Application Programming Interface (API) allowing access to the data that underpin ibTB. The development of such an API will allow third-party apps (e.g. cattle trading apps) to more readily utilise bTB data.

The authors believe that the quality of animal health research and the ability to use the data to inform farmers' biosecurity decision-making are in direct correlation to the quality and availability of the relevant data. The analysis and management of other animal health diseases would benefit if the epidemiological data were more pro-actively shared, using systems like ibTB as a template.

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Enjeux et perspectives de l'échange et du partage de données de santé animale à des fins de recherche et de gestion des maladies : un exemple concret relatif à la tuberculose bovine

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Résumé

L'échange et le partage de données sur les maladies animales sont des pratiques à encourager. En effet, l'analyse de ces données permet d'étoffer les connaissances sur les maladies animales et peut aussi apporter un nouvel éclairage sur leur gestion. Néanmoins, la nécessité de se conformer aux règles sur la protection des données pose souvent des difficultés pratiques lors des échanges de ce type de données à des fins d'analyse.

Les auteurs expliquent les difficultés rencontrées en matière d'échange de données de santé animale en Angleterre, en Écosse et au Pays de Galles (Grande-Bretagne), ainsi que les méthodes utilisées, à partir de l'exemple concret des données relatives à la tuberculose bovine. L'échange et le partage de données sont réalisés par l'Agence britannique

de santé animale et végétale, pour le compte du ministère britannique de l'Environnement, de l'Alimentation et des Affaires rurales et des gouvernements gallois et écossais. Il convient de préciser que les données de santé animale dont il s'agit sont celles conservées au niveau de la Grande-Bretagne seulement (et non du Royaume-Uni, qui inclut l'Irlande du Nord), étant donné que le ministère de l'Agriculture, de l'Environnement et des Affaires rurales de l'Irlande du Nord possède ses propres systèmes de données.
La tuberculose bovine est le principal problème de santé animale auquel sont confrontés les éleveurs de bovins en Angleterre et au Pays de Galles, et le plus coûteux à traiter. La survenue de la tuberculose bovine est une catastrophe pour les éleveurs affectés et leur communauté. En outre, le coût annuel de son contrôle s'élève à plus de 150 millions de livres pour le contribuable britannique.
Les auteurs décrivent deux méthodes d'échange et de partage de données : la première est celle où une institution de recherche demande et obtient l'accès à des données particulières afin de réaliser une étude épidémiologique ou scientifique ; la deuxième consiste à publier les données de manière proactive et constructive, en les rendant facilement accessibles. Un exemple concret de cette deuxième méthode est décrit en détail : il s'agit du site web d'information sur la tuberculose bovine (https://ibtb.co.uk), d'accès libre, qui diffuse des informations sur cette maladie à l'intention des éleveurs et des professionnels de la santé animale.

Mots-clés
Données de santé animale – Échange et partage d'informations – Gestion de maladies – ibTB – Information sur la tuberculose bovine – Règlement général sur la protection des données (Royaume-Uni) – RGPD (Royaume-Uni) – Tuberculose bovine.

Dificultades y posibilidades de la puesta en común de datos zoosanitarios con fines de investigación y gestión de enfermedades: estudio de un ejemplo referido a la tuberculosis bovina

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Resumen
Convendría alentar la puesta en común de datos zoosanitarios, pues el análisis de estos datos nos ayudará a conocer más y mejor las enfermedades animales y, a la postre, puede darnos pistas sobre la mejor manera de afrontarlas. Ocurre a menudo, sin embargo, que el prescriptivo cumplimiento de las reglas de protección de datos plantee dificultades prácticas para poner estos datos en común con fines de análisis.
Los autores, empleando como ejemplo un estudio sobre la tuberculosis bovina, describen esas dificultades y los métodos utilizados para compartir datos zoosanitarios en Inglaterra, Escocia y Gales (Gran Bretaña). En el ejemplo descrito, la Agencia de Sanidad Animal y Vegetal del Reino Unido fue la instancia que impulsó la puesta en común de los datos en nombre del Departamento de Medio Ambiente, Alimentación y Asuntos Rurales del Reino Unido y de los gobiernos galés y escocés. Conviene puntualizar que los datos zoosanitarios cubren el territorio de Gran Bretaña (y no de todo el Reino Unido, que incluye Irlanda del Norte), ya que el Departamento de Medio Ambiente, Alimentación y Asuntos Rurales norirlandés dispone de su propio sistema de datos independiente.
La tuberculosis bovina es el problema zoosanitario más importante y oneroso al que hacen frente las explotaciones de vacuno en Inglaterra y Gales. Esta enfermedad no solo puede ser devastadora para los productores y profesionales del sector, sino que la lucha contra ella cuesta al contribuyente británico más de 150 millones de libras al año.
Los autores describen dos métodos para compartir de datos: en el primero de ellos, un establecimiento universitario solicita y recibe los datos con fines de análisis científico o epidemiológico; en el segundo, una entidad toma la iniciativa de hacer públicos los datos de forma accesible y coherente. Los autores exponen en detalle un ejemplo del segundo procedimiento, a saber, el sitio web de información sobre la tuberculosis bovina (https://ibtb.co.uk) en libre acceso, en el cual se publican datos sobre la enfermedad dirigidos a los profesionales del sector pecuario y la sanidad animal.

Palabras clave

Atención de enfermedades – Datos zoosanitarios – ibTB – Información sobre la tuberculosis bovina – Puesta en común de datos – Reglamento general de protección de datos del Reino Unido – RGPD del Reino Unido – Tuberculosis bovina.

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